

Microcontroller Simulators

Simulators let you test the software for a microcontroller without using the real microcontroller. Used in early stages of program development, these programs are executed on general-purpose computers to simulate the instruction sets of microcontrollers

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The advancement in integration technology has resulted in the possibility of building an entire computing system on a single chip known as microcontroller. These chips are mainly used in embedded applications.

The microcontroller must contain a microprocessor, memory (RAM and/or ROM), input ports and output ports. Sometimes a clock is also included. If a microcontroller system is to be constructed with the program permanently stored/held in ROM, considerable testing of the program is required.

Simulators let you test the software for a microcontroller without using the real microcontroller. Used in early stages of program development, these programs are executed on general-purpose computers to simulate the instruction sets of microcontrollers. In many cases, it is possible to develop the entire project just using the simulator and then burn the compiled code into the real microcontroller. However, this method does not test the input or output devices.

This month's EFY-CD focuses on microcontroller simulator software. Here is a preview:

gpsim

gpsim is an open source simulator for Microchip's PIC microcontrollers. It supports all three families of PICs: 12-bit, 14-bit and 16-bit cores.

Distributor: Being free and open source, this software can be downloaded freely from the website of the software.

*Home Page: www.gpsim.sourceforge.net/
Download URL: <http://gpsim.sourceforge.net/#src>*

Platforms Supported: Linux, MS Windows

Real Pic Simulator

Real Pic Simulator is a professional PIC simulator for Microchip's PIC microcontrollers. The simulation process is done in real time with user interaction through different visual components. The main goal of this project is speed. The developer claims it to be the fastest PIC simulator.

Distributor: The 30-day trial software can be downloaded freely from the website. The full version can also be purchased from the website.

*Home Page: www.digitalelectrosoft.com/
Download URL: www.digitalelectrosoft.com/download-pic-simulator
Platforms Supported: MS Windows, Linux (need Wine to run Real Pic Simulator)*

PIC Development Studio

PIC Development Studio is a simulator for the PIC16F84 microcontroller. It also provides a plug-in framework, making it possible to develop custom components. A library of readymade components is included.

Distributor: Being free and open source, this software can be downloaded freely from the website.

*Home Page: www.picdev.sourceforge.net/webpage/web.php?page=main
Download URL: www.sourceforge.net/projects/picdev/files/
Platforms Supported: Linux, MS Windows*

PICsim

PICsim emulates microcontroller PIC16F628 and peripherals such as USART, EEPROM, WDT and timers. The simulator architecture permits easy

implementation of external elements in 'C' language. PicsimLab1 is a real-time emulator of a development board.

Distributor: Being free and open source, this software can be downloaded freely from the website.

*Home Page: www.sourceforge.net/projects/picsim/
Download URL: www.sourceforge.net/projects/picsim/files/picsim/picsim-0.2/picsimlab1-0.2_win.exe/download
Platform Supported: MS Windows*

Arduino

Arduino is an open-source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for designers, hobbyists and anyone interested in creating interactive objects or environments.

Distributor: Being free and open source, this software can be downloaded freely from the website.

*Home Page: www.arduino.cc
Download URL: www.arduino.cc/en/Main/Software
Platforms Supported: Linux, Mac, MS Windows*

OSHO Software

This collection from Oshonsoft consists of simulators for AVR, PIC, PIC18, PIC10F, Z80 and 8085.

Distributor: The software can be purchased online from the website. The website also offers trial version of the simulators for download.

*Home Page: www.oshonsoft.com
Download URL: www.oshonsoft.com/downloads.html
Platforms Supported: Linux, MS Windows*

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